

UNIT-III

MEDIA ACCESS CONTROL

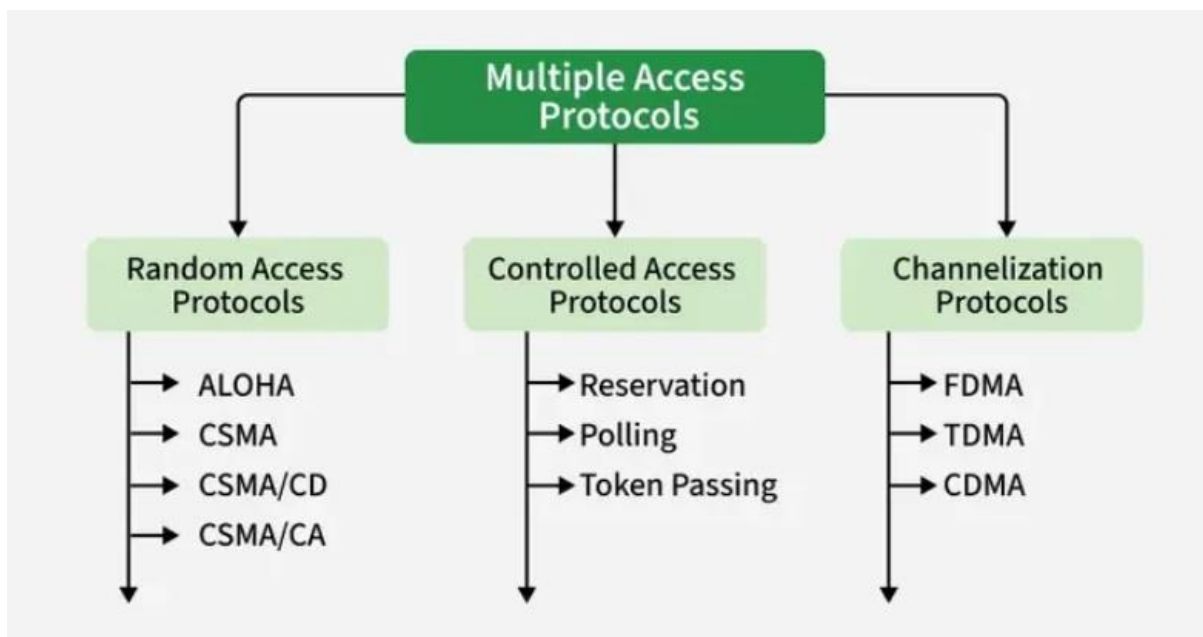
UNIT-III: Media Access Control: Random Access & The Network Layer Design Issues

Media Access Control: Random Access: ALOHA, Carrier sense multiple access (CSMA), CSMA with Collision Detection, CSMA with Collision Avoidance, **Controlled Access:** Reservation, Polling, Token Passing, **Channelization:** frequency division multiple Access(FDMA), time division multiple access(TDMA), code division multiple access(CDMA).

The Network Layer Design Issues – Store and Forward Packet Switching-Services Provided to the Transport layer- Implementation of Connectionless Service-Implementation of Connection Oriented Service- Comparison of Virtual Circuit and Datagram Networks.

Multiple Access Protocols in Computer Networks

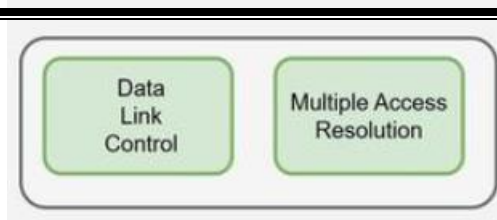
Multiple Access Protocols are a set of rules used in computer networks to control how multiple devices share and access a common communication channel. Since simultaneous transmissions by multiple devices can lead to data collisions and loss, these protocols ensure orderly, fair, and efficient use of the shared medium by coordinating when and how each device is allowed to transmit data.



Role in the OSI Model

The Data Link Layer is responsible for the transmission of data between nodes. It has two major functions:

1. **Data Link Control (DLC):** Ensures reliable transmission using framing, error control flow control (e.g., Stop-and-Wait ARQ).
2. **Multiple Access Control (MAC):** Manages access when multiple stations share a common channel (non-dedicated link).



Types of Multiple Access Protocols

Multiple Access Protocols are categorized into three major types:

- Random Access Protocols
- Controlled Access Protocols
- Channelization Protocols

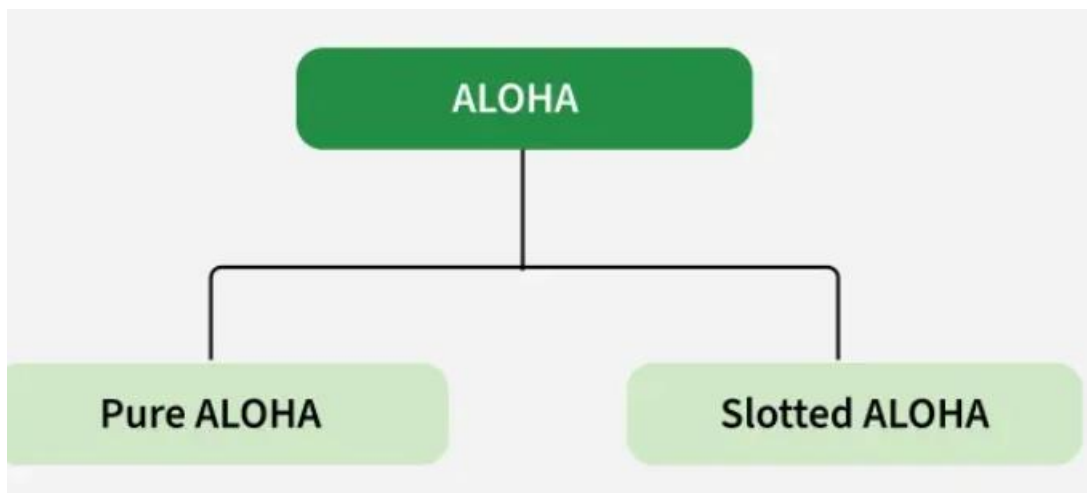
Random Access Protocols

In this approach, all stations have equal priority. Transmission decisions are based on the state of the channel (idle or busy). Below are its Features:

- No fixed order of transmission.
- No fixed time slot assignment.

1. ALOHA

- Designed for wireless LANs and shared media.
- Multiple stations can transmit simultaneously, leading to collisions.



Types of Aloha:

- Pure ALOHA
- Slotted ALOHA

Aloha is a type of Random access protocol it was developed at the University of Hawaii in early 1970, it is a LAN-based protocol this type there are more chances of occurrence of collisions during the transmission of data from any source to the destination, Aloha has two types one Pure Aloha and another Slotted Aloha.

Pure Aloha

Pure Aloha can be termed as the main Aloha or the original Aloha. Whenever any frame is available, each station sends it, and due to the presence of only one channel for communication, it can lead to the chance of collision.

In the case of the pure aloha, the user transmits the frame and waits till the receiver acknowledges it, if the receiver does not send the acknowledgment, the sender will assume that it has not been received and sender resends the acknowledgment.

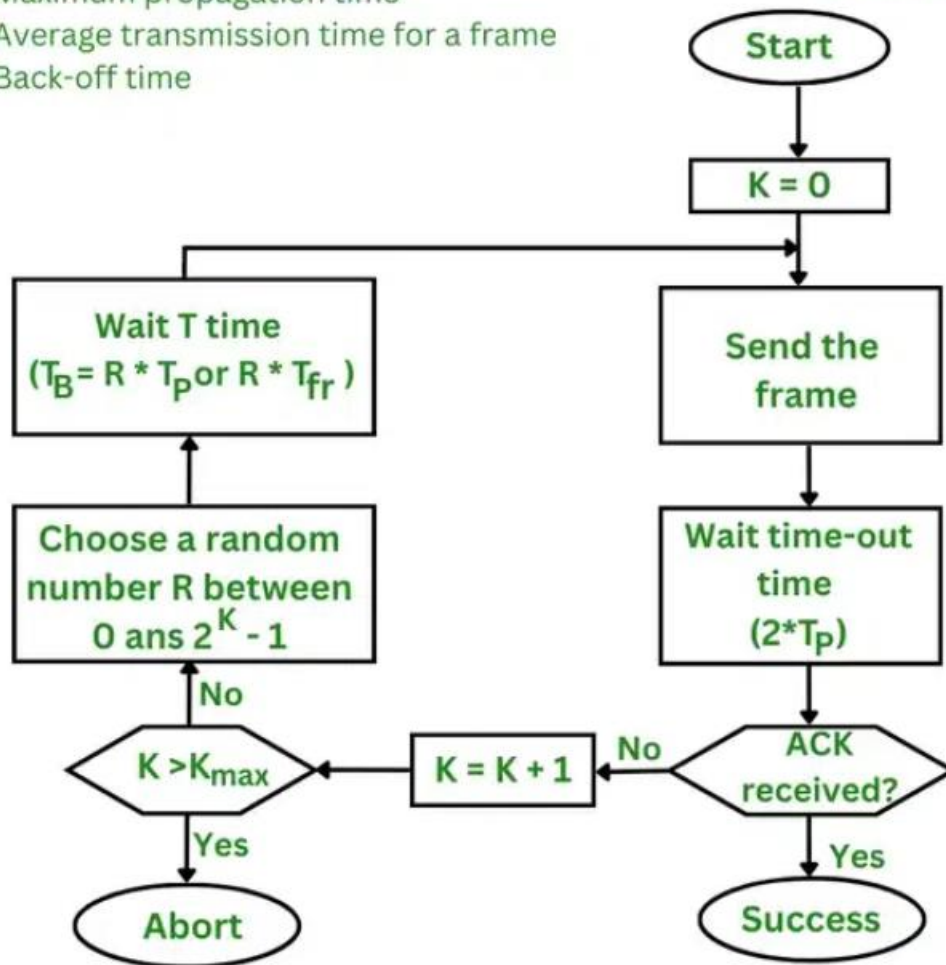
K : Number of attempts

T_p : Maximum propagation time

T_{fr} : Average transmission time for a frame

T_B : Back-off time

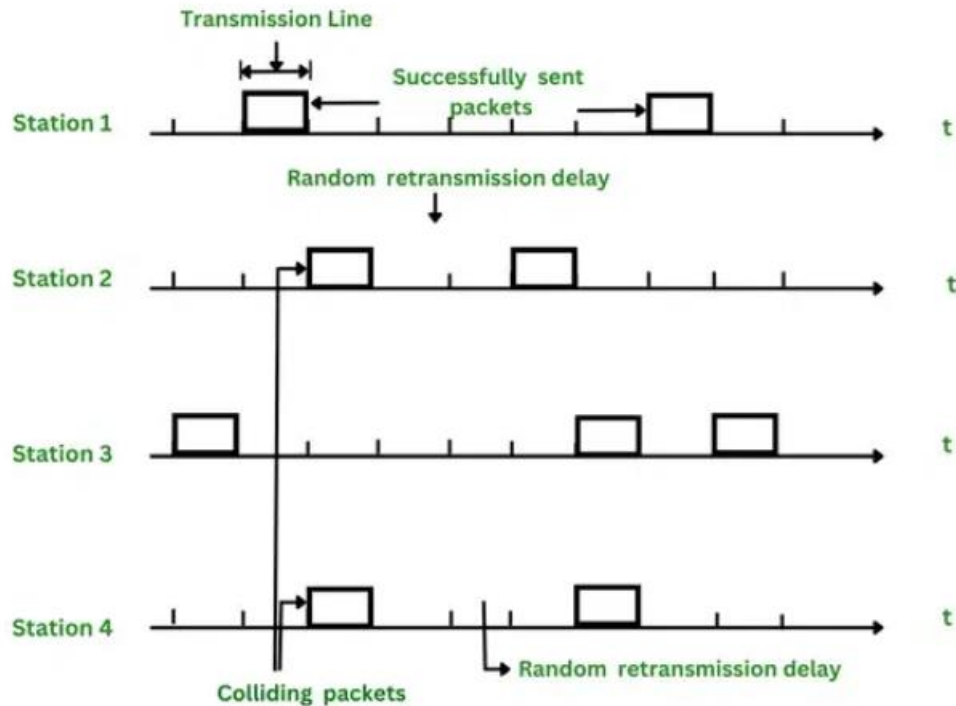
Station has a
frame to send



Slotted Aloha

Slotted Aloha is simply an advanced version of pure Aloha that helps in improving the communication network. A station is required to wait for the beginning of the next slot to transmit. The vulnerable period is halved as opposed to Pure Aloha.

Slotted Aloha helps in reducing the number of collisions by properly utilizing the channel and this basically results in the somehow delay of the users. In Slotted Aloha, the channel time is separated into particular time slots.



Differences Between Pure Aloha and Slotted Aloha

The difference between Pure and Slotted Aloha lies in their approach to handling data collisions. While Pure Aloha sends data anytime, Slotted Aloha reduces collisions by organizing time slots.

Pure Aloha	Slotted Aloha
In this Aloha, any station can transmit the data at any time.	In this, any station can transmit the data at the beginning of any time slot.
In this, The time is continuous and not globally synchronized.	In this, the time is discrete and globally synchronized.
Vulnerable time for Pure Aloha = $2 \times T_t$	Vulnerable time for Slotted Aloha = T_t
In Pure Aloha, the Probability of successful transmission of the data packet = $G \times e^{-2G}$	In Slotted Aloha, the Probability of successful transmission of the data packet = $G \times e^{-G}$
In Pure Aloha, Maximum efficiency = 18.4%	In Slotted Aloha, Maximum efficiency = 36.8%

Pure Aloha

Pure Aloha doesn't reduce the number of collisions to half.

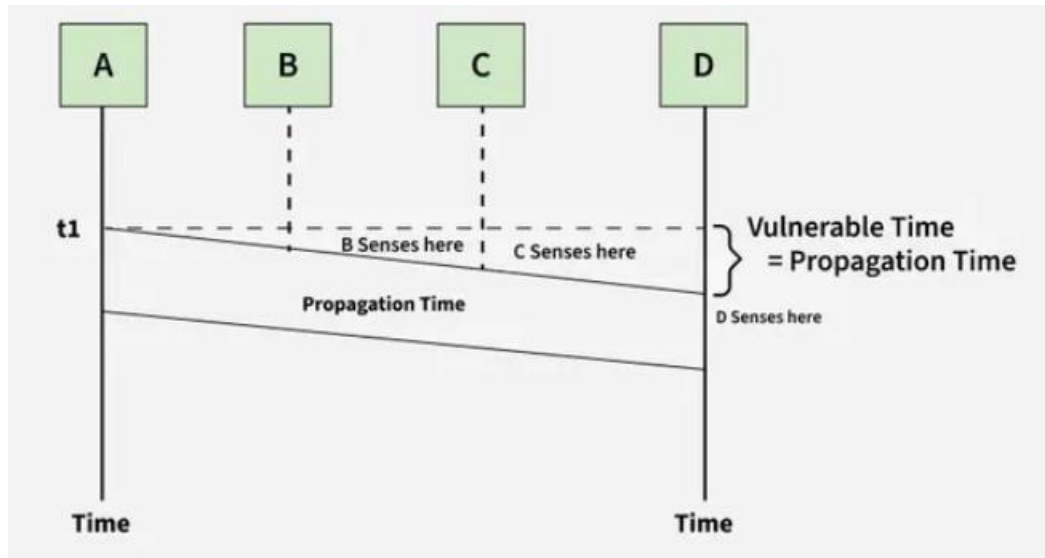
Slotted Aloha

Slotted Aloha reduces the number of collisions to half and doubles the efficiency of Pure Aloha.

2. CSMA (Carrier Sense Multiple Access)

Before transmitting, stations sense the medium:

- If idle -> transmit.
- If busy -> wait.



CSMA

Types of CSMA:

- CSMA/CD (Collision Detection)
- CSMA/CA (Collision Avoidance)

Controlled Access Protocols

Here, devices take turns accessing the channel, preventing collisions. For example:

- **Reservation:** Stations reserve slots before transmission.
- **Polling:** Central controller polls each device in turn.
- **Token Passing:** A token (special packet) is passed; only the device holding the token can transmit.

Channelization Protocols

Bandwidth is divided and allocated so that multiple users can transmit simultaneously.

1. Frequency Division Multiple Access (FDMA)

- Bandwidth is divided into frequency bands.
- Guard bands prevent overlap.

2. Time Division Multiple Access (TDMA)

- Time is divided into slots.
- Each station transmits only in its assigned slot.
- Needs synchronization.

3. Code Division Multiple Access (CDMA)

- All stations transmit simultaneously using unique codes.
- Widely used in mobile communication.

4. Orthogonal Frequency Division Multiple Access (OFDMA)

- Bandwidth divided into small subcarriers.
- Widely used in 4G/5G for high data rates.
- High throughput, supports multimedia.
- Complex design, high power requirements.

5. Spatial Division Multiple Access (SDMA)

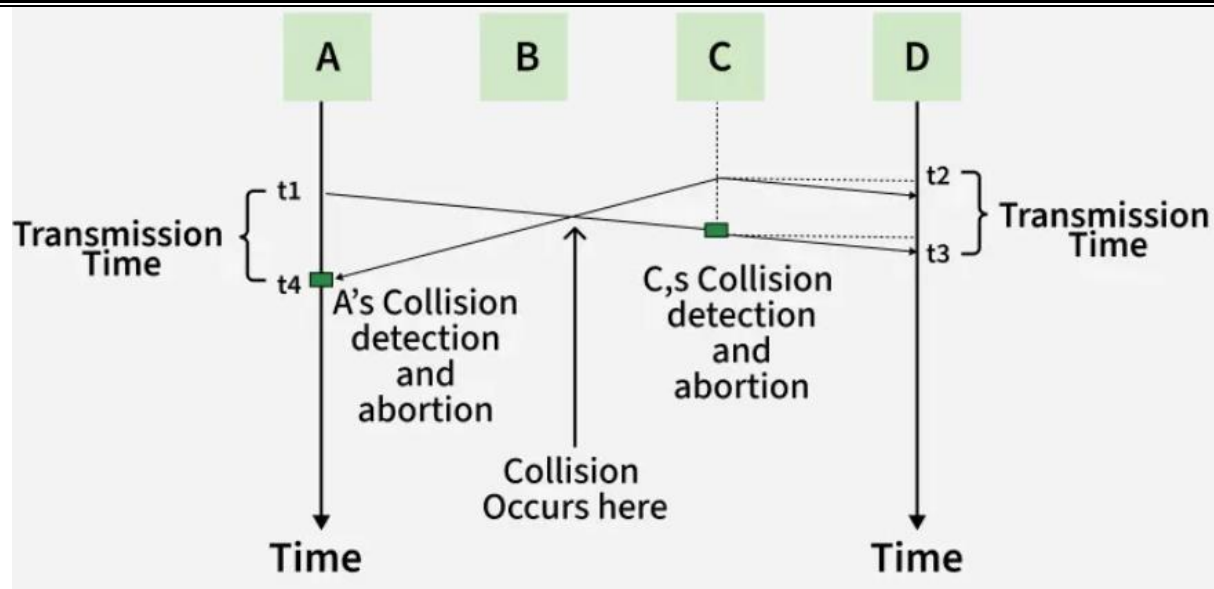
- Uses multiple antennas (MIMO).
- Separates users spatially.
- Improves data rate and signal quality.

Features of Multiple Access Protocols

- **Contention-Based Access:** Devices compete for channel.
- **Carrier Sensing:** Medium is sensed before transmitting.
- **Collision Handling:** Detection (CSMA/CD) or avoidance (CSMA/CA).
- **Token Passing:** Sequential access control.
- **Bandwidth Utilization:** Varies depending on protocol efficiency.

➤ Collision Detection in CSMA/CD

CSMA/CD (Carrier Sense Multiple Access/ Collision Detection) is a media access control method that was widely used in Early Ethernet technology/LANs, when there used to be shared Bus Topology and each node (Computers) was connected by Coaxial Cables.



How Does CSMA/CD Work?

Step 1: Check if the sender is ready to transmit data packets.

Step 2:

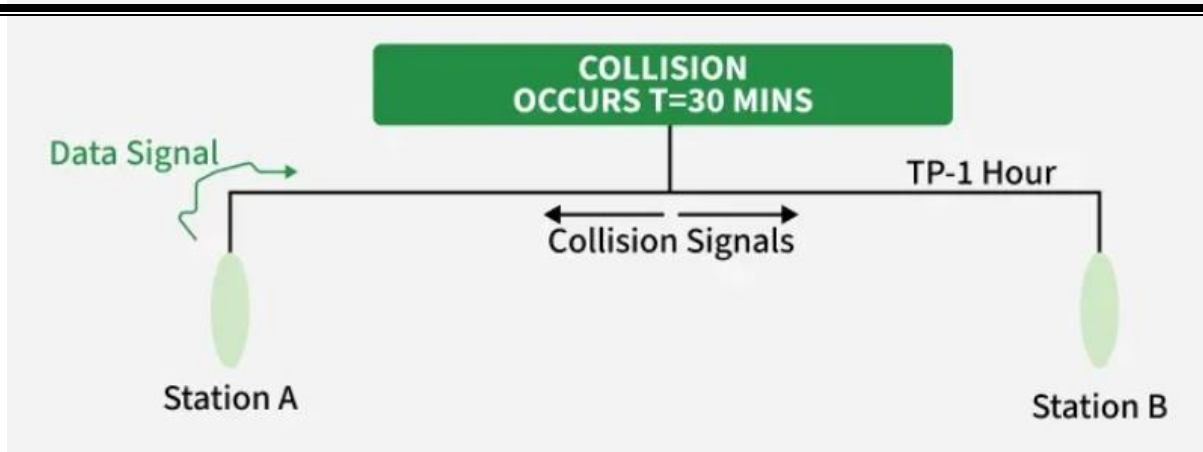
- Check if the transmission link is idle.
- The sender has to keep on checking if the transmission link/medium is idle.
- For this, it continuously senses transmissions from other nodes.
- The sender sends dummy data on the link. If it does not receive any collision signal, this means the link is idle at the moment.
- If it senses that the carrier is free and there are no collisions, it sends the data. Otherwise, it refrains from sending data.

Step 3:

- Transmit the data & check for collisions.
- The sender transmits its data on the link.
- CSMA/CD does not use an 'acknowledgment' system. It checks for successful and unsuccessful transmissions through collision signals.
- During transmission, if a collision signal is received by the node, transmission is stopped.
- The station then transmits a jam signal onto the link and waits for random time intervals before it resends the frame.
- After some random time, it again attempts to transfer the data and repeats the above process.

Step 4: If no collision was detected in propagation, the sender completes its frame transmission and resets the counters.

How Does a Station Know if Its Data Collide?

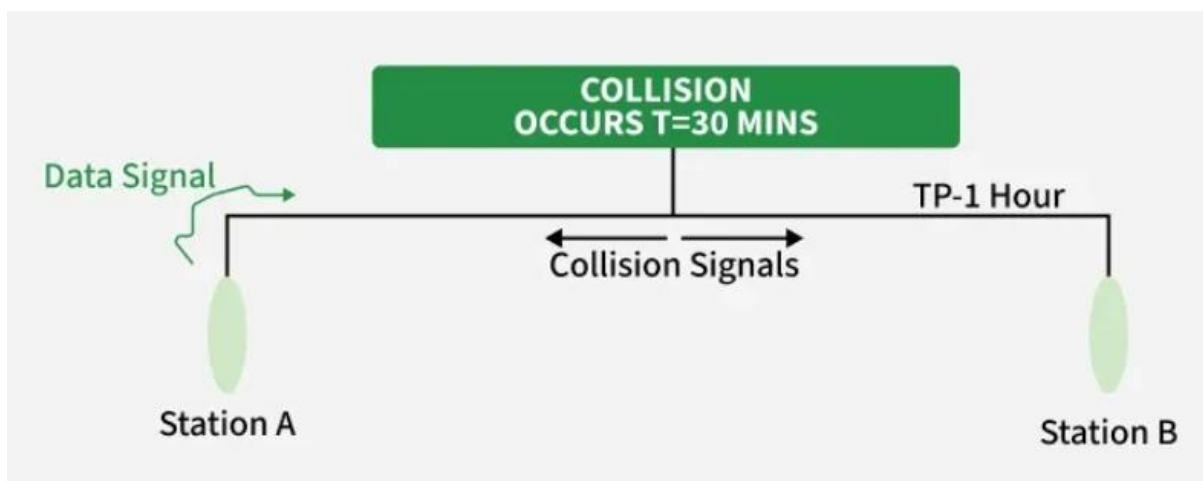


Consider the above situation, Two stations, A & B:

- Propagation Time: $T_p = 1\text{hr}$ (Signal takes 1 hr to go from A to B)
- At time $t=0$, A transmits its data, $t=30\text{ mins}$ -> Collision occurs
- After the collision occurs, a collision signal is generated and sent to both A & B to inform the stations about the collision.
- Since the collision happened midway, the collision signal also takes 30 minutes to reach A & B.
- Therefore, $t=1\text{hr}$ -> A & B receive collision signals. This collision signal is received by all the stations on that link.

How to Ensure that it is our Station's Data that Collided?

For this, Transmission time (T_t) > Propagation Time (T_p) [Rough bound], This is because we want that before we transmit the last bit of our data from our station, we should at least be sure that some of the bits have already reached their destination. This ensures that the link is not busy and collisions will not occur.



At time $t=0$, A transmits its data, $t=50:59\text{ mins}$: Collision occurs

- This collision occurs just before the data reaches B. Now the collision signal takes 50:59 minutes again to reach A.
- Hence, A receives the collision information approximately after 2 hours, that is, after $2 \times T_p$.
- Hence, to ensure tighter bound, to detect the collision completely, $T_t \geq 2 \times T_p$.

- This is the maximum collision time that a system can take to detect if the collision was of its own data.

What should be the Minimum length of the Packet to be Transmitted?

Transmission Time = T_t = Length of the packet/ Bandwidth of the link. Substituting above, we get,
 Length of the packet/ Bandwidth of the link $\geq 2 * T_p$

Length of the packet $\geq 2 * T_p * \text{Bandwidth of the link}$

Length of the packet $\geq 2 * T_p * \text{Bandwidth of the link}$

Note: Padding helps in cases where we do not have such long packets. We can pad extra characters to the end of our data to satisfy the above condition.

Features of CSMA/CD

- **Carrier Sense:** Before transmitting data, a device listens to the network to check if the transmission medium is free. If the medium is busy, the device waits until it becomes free before transmitting data.
- **Multiple Access:** In a CSMA/CD network, multiple devices share the same transmission medium. Each device has equal access to the medium and any device can transmit data when the medium is free.
- **Collision Detection:** If two or more devices transmit data simultaneously, a collision occurs. When a device detects a collision, it immediately stops transmitting and sends a jam signal to inform all other devices on the network of the collision. The devices then wait for a random time before attempting to transmit again, to reduce the chances of another collision.
- **Backoff Algorithm:** In CSMA/CD, a backoff algorithm is used to determine when a device can retransmit data after a collision. The algorithm uses a random delay before a device retransmits data, to reduce the likelihood of another collision occurring.
- **Minimum Frame Size:** CSMA/CD requires a minimum frame size to ensure that all devices have enough time to detect a collision before the transmission ends. If a frame is too short, a device may not detect a collision and continue transmitting, leading to data corruption on the network.

Limitations of CSMA/CD

- **Limited Scalability:** CSMA/CD has limitations in terms of scalability and it may not be suitable for large networks with a high number of devices.
- **Vulnerability to Collisions:** While CSMA/CD can detect collisions, it cannot prevent them from occurring. Collisions can lead to data corruption, retransmission delays and reduced network performance.
- **Inefficient Use of Bandwidth:** CSMA/CD uses a random [backoff algorithm](#) that can result in inefficient use of network [bandwidth](#) if a device continually experiences collisions.
- **Susceptibility to Security Attacks:** CSMA/CD does not provide any security features and the protocol is vulnerable to security attacks such as packet sniffing and [spoofing](#).

➤ CSMA with Collision Avoidance (CSMA/CA)

Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) is a network protocol for carrier transmission that operates in the Medium Access Control (MAC) layer. In contrast to CSMA/CD (Carrier Sense Multiple Access/Collision Detection) that deals with collisions after their occurrence, CSMA/CA prevents collisions prior to their occurrence.

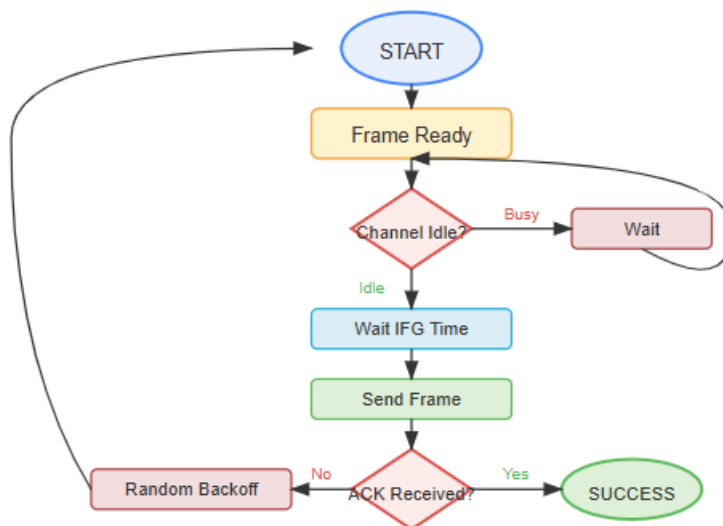
CSMA/CA is primarily used in wireless networks where collision detection is difficult or impossible due to the hidden terminal problem and the inability to simultaneously transmit and receive on the same channel.

How CSMA/CA Works

The algorithm of CSMA/CA follows these steps:

- When a frame is ready, the transmitting station checks whether the channel is idle or busy.
- If the channel is busy, the station waits until the channel becomes idle.
- If the channel is idle, the station waits for an **Inter-frame Gap (IFG)** amount of time and then sends the frame.
- After sending the frame, it sets a timer and waits for acknowledgement from the receiver.
- If it receives the acknowledgement before timer expiry, it marks a successful transmission.
- Otherwise, it assumes a collision occurred, waits for a random **back-off time** period, and restarts the algorithm.

CSMA/CA Algorithm Flowchart



CSMA/CA vs CSMA/CD

Feature	CSMA/CA	CSMA/CD
Collision Handling	Prevention (before collision)	Detection (after collision)
Primary Use	Wireless networks (WiFi)	Wired networks (Ethernet)
Acknowledgment	Required	Not required
Efficiency	Lower (due to waiting times)	Higher (immediate detection)

Advantages of CSMA/CA

- CSMA/CA prevents collisions proactively, reducing data loss.
- Due to acknowledgments, reliable data transmission is ensured.

- It avoids wasteful retransmissions by preventing collisions.
 - Well-suited for wireless networks where collision detection is difficult.
- Disadvantages of CSMA/CA
- The algorithm requires longer waiting times, reducing overall throughput.
 - Higher power consumption due to constant channel sensing and acknowledgment overhead.
 - More complex implementation compared to CSMA/CD.

➤ Controlled Access Protocols in Computer Network

In computer networks, Controlled Access Protocols (CAPs) manage how multiple devices share a common communication medium. These protocols control the right to transmit in such a way that collisions are avoided, efficiency is maintained, and fair access is ensured.

Note: Unlike random access methods (e.g., ALOHA, CSMA), in controlled access, stations do not transmit at will. Instead, they follow a systematic method to determine which station has the right to send at a given time.

What is Controlled Access?

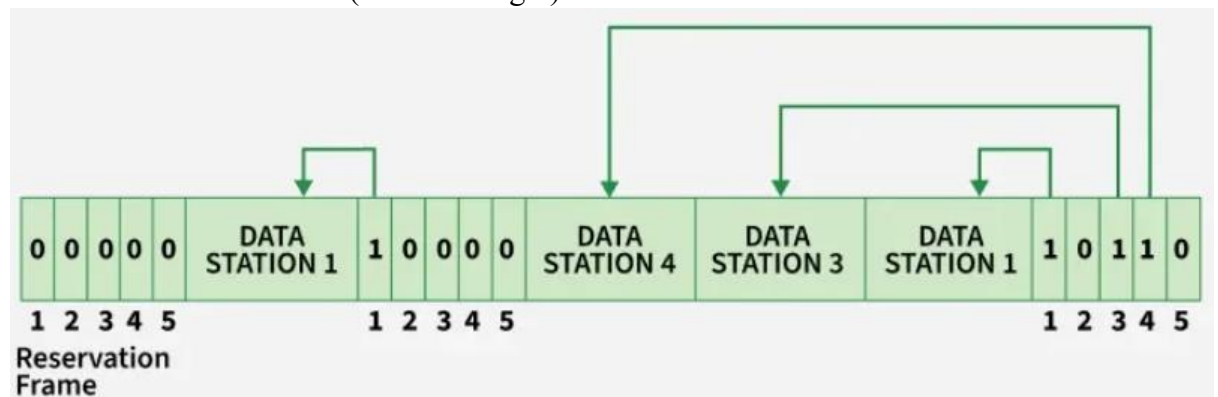
In Controlled Access, before transmitting data, a station must first obtain permission to access the channel. At any given time, only one node can send data, preventing collisions. This coordination is achieved using one of three methods:

- Reservation
- Polling
- Token Passing

Reservation

In this method, stations reserve slots in advance before transmission. The timeline has two parts:

- Reservation Interval (fixed length): divided into slots, one per station.
- Data Transmission Period (variable length): reserved stations send data in order.



Example: If there are M stations, the reservation interval is divided into M slots. Each station signals intent by inserting a bit into its slot. Then, only the reserved stations transmit in the data phase.

Advantages

- Predictable access time and throughput.
- Priorities can be set for faster access.
- Supports QoS (Quality of Service) for applications like real-time audio/video.
- Efficient bandwidth utilization.

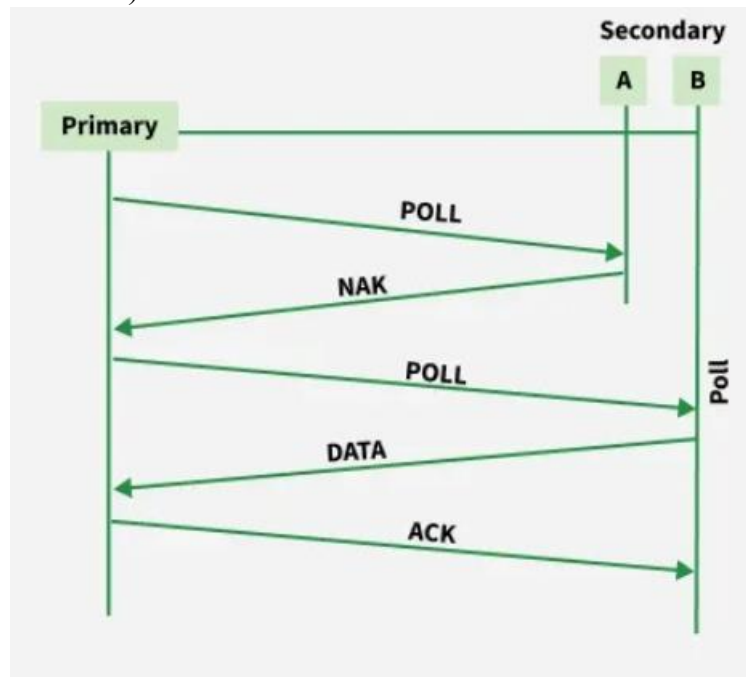
Disadvantages

- Performance decreases under light loads (wasted slots).
- High dependency on controlled synchronization.

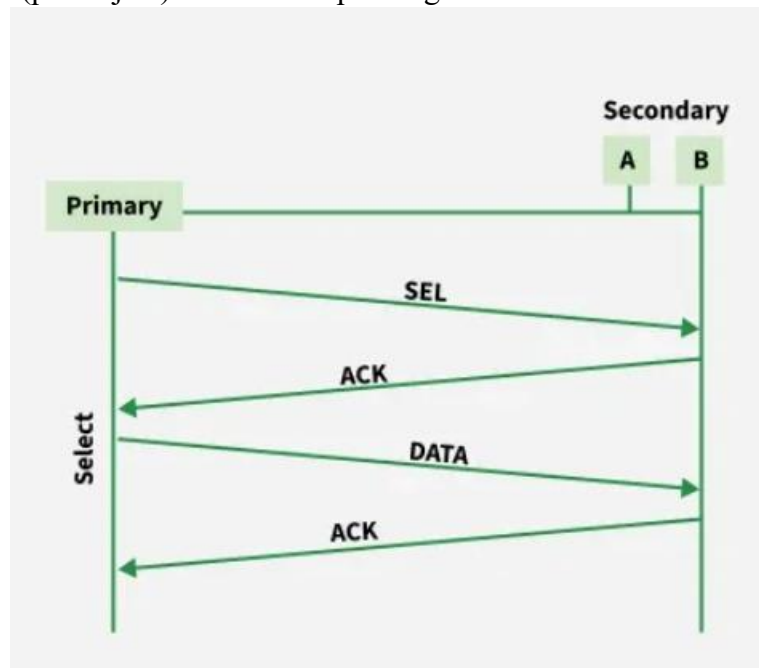
Polling

Here, a controller (primary station) polls each node (secondary station) in sequence, granting permission to send. The controller sends a message with the address of a station. The addressed station responds:

- Sends data (if available)



- Sends a NAK (poll reject) if no data is pending.



Advantages

- Predictable access times and bandwidth.
- High efficiency, since no slot is wasted.
- Priorities can be assigned.

Disadvantages

- Extra overhead due to polling messages.
- High dependence on controller reliability.
- Turnaround time increases under light loads.

- Efficiency formula

$$\eta = Tt + T_{poll}Tt$$

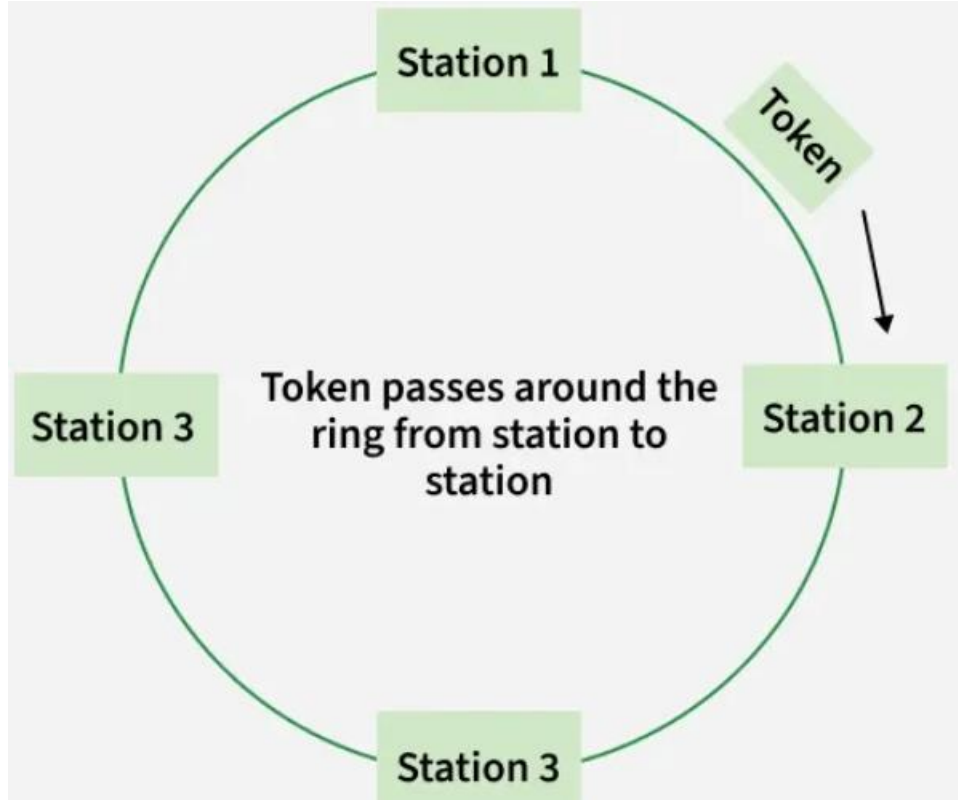
where:

- $T_t = T_t$ = data transmission time
- $T_{poll} = T_{poll}$ = polling overhead time

Token Passing

In Token Passing, stations are logically connected (ring or bus). A special frame called a token circulates among stations:

- The token represents permission to transmit.
- A station can send data only when it holds the token.
- If no data is ready, the token is simply passed on.



Common Implementations

- **Token Ring** – token passed in a ring topology.
- **Token Bus** – stations use a bus medium but pass the token in a predefined logical order.

Advantages

- Excellent performance under high loads.
- Built-in error recovery and debugging features.
- Predictable throughput.

Disadvantages

- Expensive setup and hardware.
- Token management issues (lost/duplicate tokens, station failures).
- Complex implementation.

Performance Parameters

- Delay: Average delay = $a/N = a/N$, where $a = T_p/T_t = T_p/T_t$.
- Throughput:
 1. For $a < 1$: $S = 1/(1 + a/N)$
 2. For $a > 1$: $S = a/(1 + a/N)$

where:

- N = number of stations
- T_p = propagation delay
- T_t = transmission delay

Comparison of Controlled Access Methods

Feature	Reservation	Polling	Token Passing
Access Control	Slot-based reservation	Central controller polls	Token circulation
Efficiency	High (under heavy load)	Moderate (polling overhead)	High (predictable)
Collision Chance	None	None	None
Overhead	Reservation slots	Polling messages	Token management
Best For	Real-time multimedia	Small/medium networks	Large/high-load networks

➤ TDMA - Technology

Time Division Multiple Access (TDMA) is a digital cellular telephone communication technology. It facilitates many users to share the same frequency without interference. Its technology divides a signal into different timeslots, and increases the data carrying capacity.

Time Division Multiple Access (TDMA) is a complex technology, because it requires an accurate synchronization between the transmitter and the receiver. TDMA is used in digital mobile radio systems. The individual mobile stations cyclically assign a frequency for the exclusive use of a time interval.

In most of the cases, the entire system bandwidth for an interval of time is not assigned to a station. However, the frequency of the system is divided into sub-bands, and TDMA is used for the multiple access in each sub-band. Sub-bands are known as **carrier frequencies**. The mobile system that uses this technique is referred as the **multi-carrier systems**.

In the following example, the frequency band has been shared by three users. Each user is assigned definite **timeslots** to send and receive data. In this example, user **B** sends

after user A, and user C sends thereafter. In this way, the peak power becomes a problem and larger by the burst communication.

FDMA and TDMA

This is a multi-carrier TDMA system. A 25 MHz frequency range holds 124 single channels (carrier frequencies 200) bandwidth of each kHz; each of these frequency channels contains 8 TDMA conversation channels. Thus, the sequence of timeslots and frequencies assigned to a mobile station is the physical channels of a TDMA system. In each timeslot, the mobile station transmits a data packet.

The period of time assigned to a timeslot for a mobile station also determines the number of TDMA channels on a carrier frequency. The period of timeslots are combined in a so-called TDMA frame. TDMA signal transmitted on a carrier frequency usually requires more bandwidth than FDMA signal. Due to the use of multiple times, the gross data rate should be even higher.

Advantages of TDMA

Here is a list of few notable advantages of TDMA –

- Permits flexible rates (i.e. several slots can be assigned to a user, for example, each time interval translates 32Kbps, a user is assigned two 64 Kbps slots per frame).
- Can withstand gusty or variable bit rate traffic. Number of slots allocated to a user can be changed frame by frame (for example, two slots in the frame 1, three slots in the frame 2, one slot in the frame 3, frame 0 of the notches 4, etc.).
- No guard band required for the wideband system.
- No narrowband filter required for the wideband system.

Disadvantages of TDMA

The disadvantages of TDMA are as follow –

- High data rates of broadband systems require complex equalization.
- Due to the burst mode, a large number of additional bits are required for synchronization and supervision.
- Call time is needed in each slot to accommodate time to inaccuracies (due to clock instability).
- Electronics operating at high bit rates increase energy consumption.
- Complex signal processing is required to synchronize within short slots.

➤ **Channelization Protocols**

Channelization is the process by which we coordinate the transmission of data in non-dedicated channels to prevent collision and crosstalk, and channelization protocols are the MAC protocols we employ to coordinate this transmission.

Channels are communication resources we assign to a user to establish communication with other users in the network. These resources include frequency, time, code, space, [bandwidth](#), and so on. We name each channelization protocol based on the kind of channel resources we employ for channelization. These protocols are:

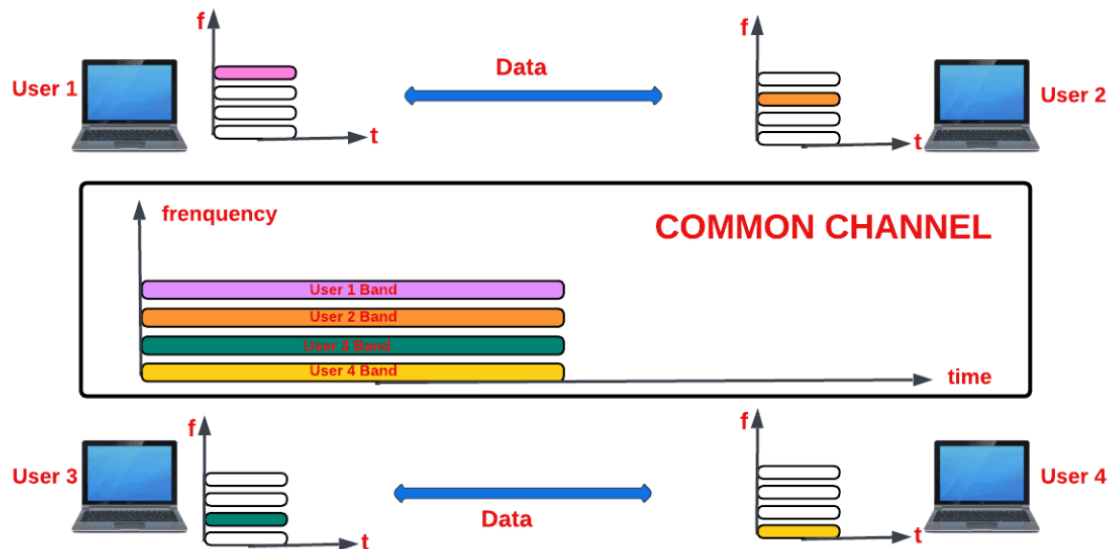
- Frequency division multiple access (FDMA)
- Time division multiple access (TDMA)
- Code division multiple access (CDMA)
- Space division multiple access (SDMA)
- Orthogonal Frequency division multiple access (OFDMA)

➤ **Frequency Division Multiple Access (FDMA) Protocol**

The FDMA protocol by utilizing the channel resource of frequency. By dividing the channel's bandwidth into non-overlapping frequency bands or slices, we allocate each

user a distinct band for exclusive transmission and reception. This enables the concurrent operation of multiple users within a single communication channel. To mitigate interference and crosstalk during data transmission, we equip each user with a customized bandpass filter complemented by a small guard band.

The following figure depicts the FDMA protocol scenario:



As seen in the above diagram, four users are attempting to access the same channel. To accommodate the four users without interfering with one another, we utilize FDMA to split the channel into four frequency bands.

Merits and Limitations of FDMA Protocols

Some notable merits which FDMA protocols offer us are:

- enable simultaneous transmission through the same frequency channel, with each user operating at a single frequency
- using frequency as the channel resource eliminates timing-related issues often encountered in TDMA protocols
- FDMA effectively resolves the near-far problem inherent in CDMA protocols by implementing bandpass filtering
- as we assign users frequency slots, this allows for distinct transmission and reception at different frequencies

However, the FDMA protocol does come with two significant limitations:

- the design of high-performing hardware filters proves to be a challenging task
- despite the inclusion of guard bands, FDMA remains susceptible to crosstalk, potentially leading to interference and transmission disruptions

Frequency Allocation Schemes in FDMA Protocols

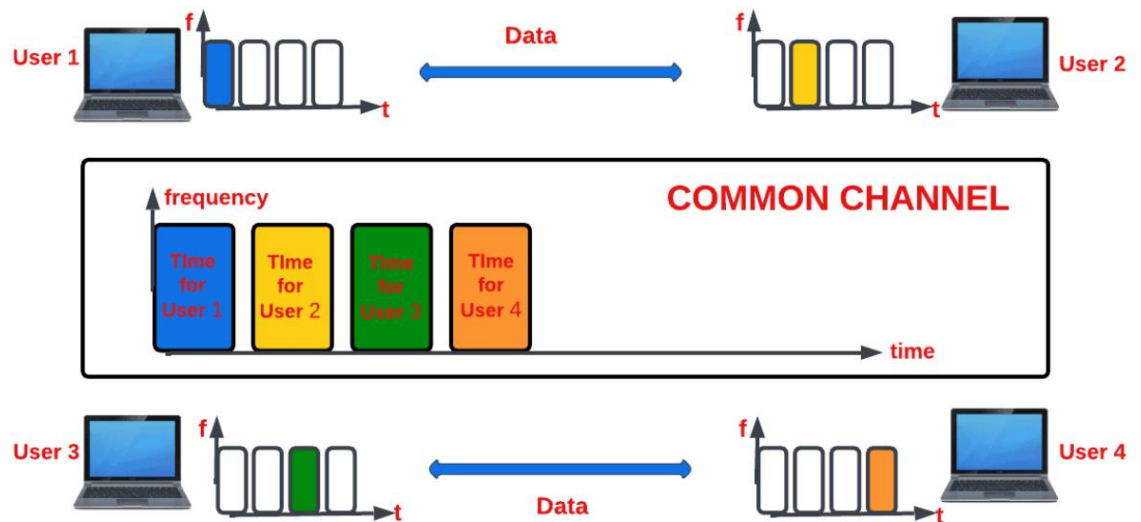
Regarding frequency slot assignment within the FDMA protocol, we employ two common methods, which are:

- Fixed assignment multiple access (FAMA)
- Demand assignment multiple access (DAMA)

We use FAMA to allocate frequency slots to users whether they utilize it or not, ensuring a fixed allocation scheme. On the other hand, we use DAMA to assign frequency slots to users only upon their explicit request, offering a more dynamic and efficient resource allocation approach.

Time Division Multiple Access (TDMA) Protocol

Time serves as the channel resource for this protocol. Using the TDMA protocol, we divide each user's signal into a different time slot and assign each time slot to a different user. This technique enables numerous users to share and use the same frequency channel. The global system of mobile communication (GSM) system, a 2G cellular system, is one significant application where we have used the TDMA protocol.



Synchronizing the various users of the channel is one of the biggest challenges we face when utilizing the TDMA protocol. To achieve synchronization, we ensure each user knows their respective slots' start and endpoints. This is difficult to achieve, especially if the users are spread over a large area. To minimize this challenge, we insert guard times to each user channel.

Another significant drawback we encounter when utilizing TDMA systems is the potential for causing interference at a frequency that closely adjoins another user's active time slot. This interference manifests as the buzzing sound we occasionally hear when a TDMA phone is near a radio or speakers.

Just like in the FDMA protocol, there are two methods we use to administer time slots to users:

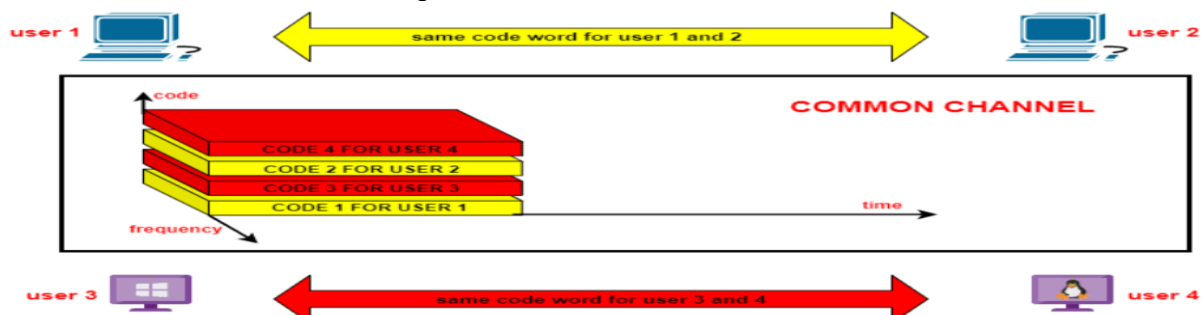
- Fixed time division multiple access (F-TDMA)
- Dynamic time division multiple access (D-TDMA)

While we use F-TDMA for fixed allocation of time slots to users, we use D-TDMA to assign time slots to users based on the traffic demand of the channel.

Code Division Multiple Access (CDMA) Protocol

The channel resource for this protocol is code. In the CDMA protocol, we allow multiple users to simultaneously transmit their data signals over the entire bandwidth of the common channel by assigning unique spreading codes to each user. This technique helps us utilize the channel more effectively.

Let's see what a CDMA protocol looks like:



The beauty of this protocol is that we don't share the frequency as we do in FDMA, and neither do we share time as we do in TDMA. In CDMA, we employ both time and frequency for users to send their data over the entire bandwidth of the channel with a unique spreading code allotted to the user. Only receivers having this unique code can receive the sent messages.

In CDMA, we use the spread spectrum principle to utilize unique transmission codes. Let us discuss this principle more in-depth.

➤ Spread Spectrum

Spread spectrum is a technique that enables us to transfer signals via communication channels by purposefully extending the signal's bandwidth beyond what is necessary for transmission. We expand the signal bandwidth in order to keep transmission interference to a bare minimum. To offset the waste of bandwidth due to overspreading, we allow multiple users to access the channel.

We implement the spread spectrum in two ways:

- Direct sequence spread spectrum (DSSS)
- Frequency hopping spread spectrum (FHSS)

Let's now discuss each method.

In DSSS, we multiply the message signal with a locally generated spreading pseudorandom noise (PN) code, which we refer to as chips, to widen the signal's bandwidth. By multiplying the signal, we broadcast at the receiver by a locally produced copy of the PN code sequence, and we can decode the information. We have incorporated DSSS into GPS and Wi-Fi networks.

To spread the message signal in FHSS, we transmit it with rapidly changing carrier frequencies among several frequencies occupying a large bandwidth. We control the "frequency hopping" of the message signal by a PN code known to the sender and receiver. In this way, we send and receive data without interference and eavesdropping.

We employ two strategies to manage CDMA protocols based on spread spectrum technologies:

- Direct sequence code division multiple access (DS-CDMA)
- Frequency hopping code division multiple access (FH-CDMA)

Based on the same logic as their spread spectrum counterparts, we apply these techniques.

Advantages and Disadvantages of CDMA

CDMA offers several advantages over other communication technologies, such as FDMA and TDMA. Firstly, CDMA significantly increases user capacity compared to FDMA and TDMA. Moreover, it allows for simultaneous voice and data communication, which is impossible with TDMA and FDMA.

Another significant advantage of CDMA is its high level of security in transmitting information. In CDMA, we employ a unique code for each user, making accessing information difficult for unauthorized users. Additionally, the probability of interference is minimal in CDMA, providing reliable communication.

Furthermore, CDMA enables us to use the entire spectrum, offering more flexibility in communication. With CDMA, we can take advantage of the entire bandwidth available, which is impossible with FDMA or TDMA.

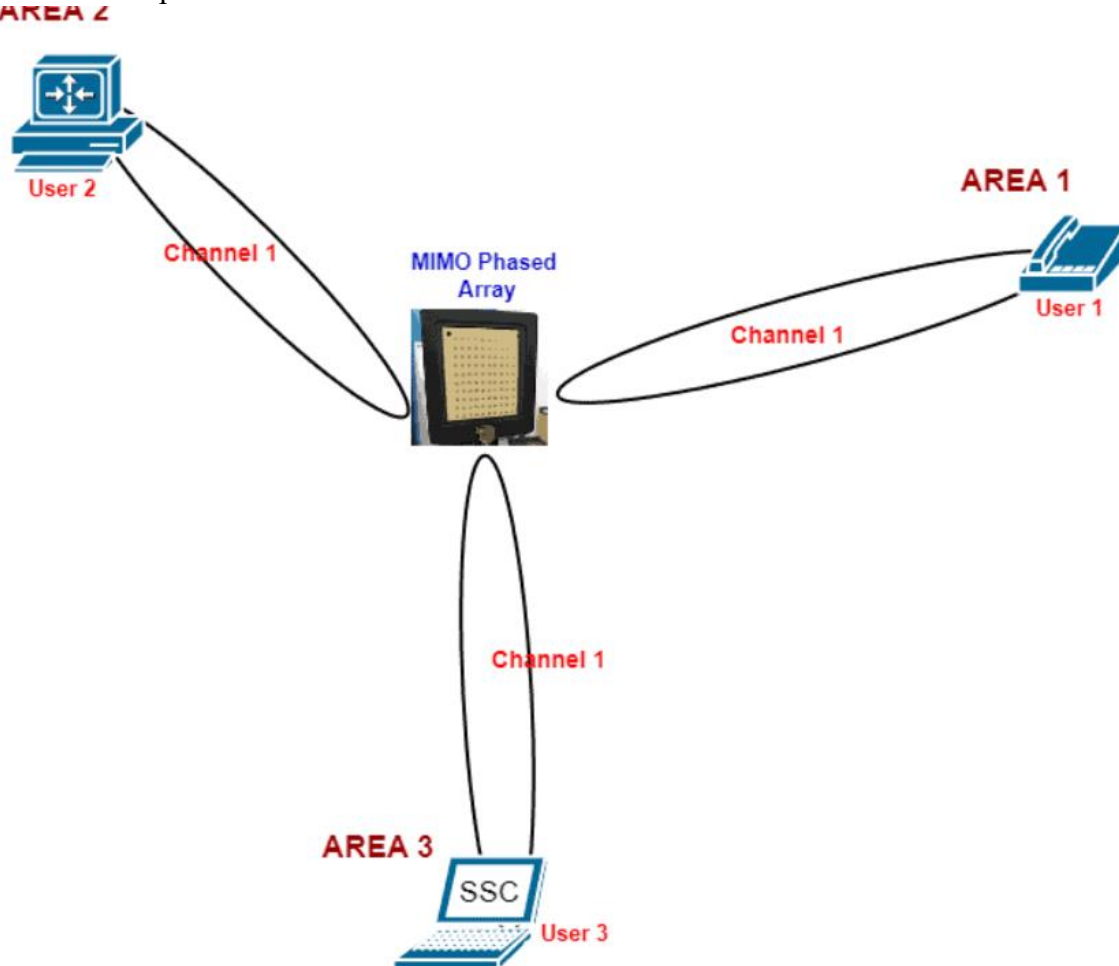
Despite its advantages, CDMA also has some significant drawbacks that must be considered. Firstly, the system's operation is highly complex, making it difficult to set up and maintain. Moreover, In CDMA, we use orthogonal codes for both the sender and the receiver in a synchronous manner to recover the original data accurately. If this requirement is unmet, the system experiences self-jamming, making communication challenging.

Space Division Multiple Access (SDMA)

In this protocol, we utilize space as the channel resource. **We allocate the same frequency to users that are separated by a certain distance, which we refer to as the reuse distance.** To accomplish this, we employ a smart and highly directional antenna that directs a beam of radio waves to a specific location in space where a user is located.

To enable the reuse of the same channel, we use another smart and highly directional antenna to beam his signal to a different location in the same space where other users are located. This smart antenna system is known as a Multiple Input Multiple Output (MIMO) system.

A typical SDMA protocol looks like this:



From the figure, we see that the same channel-1 is reused to send data to three different users because a reuse distance separates them.

By using a highly directional beam, we can minimize interference and improve the quality of communication for users in the network. Additionally, the MIMO system can increase the network's capacity by allowing multiple users to communicate simultaneously without affecting each other's signals.

This protocol provides us with some merits:

- it helps us to optimize the use of power and minimize the wastage of energy
- we are able to save the channel bandwidth and utilize it more effectively
- it helps us reduce interference to near zero
- we can control and regulate the energy we radiate to the receiver

To achieve the full potential of the SDMA protocol, we must carefully design it, which we always find very tedious.

➤ Orthogonal Frequency Division Multiple Access (OFDMA)

As the world is witnessing an increasing demand for individual data consumption, we recognize the necessity to enhance the existing protocol standards to meet this demand. To address this, we devised a solution that merged the benefits of FDMA and TDMA protocols in an orthogonal fashion, thereby preventing interference.

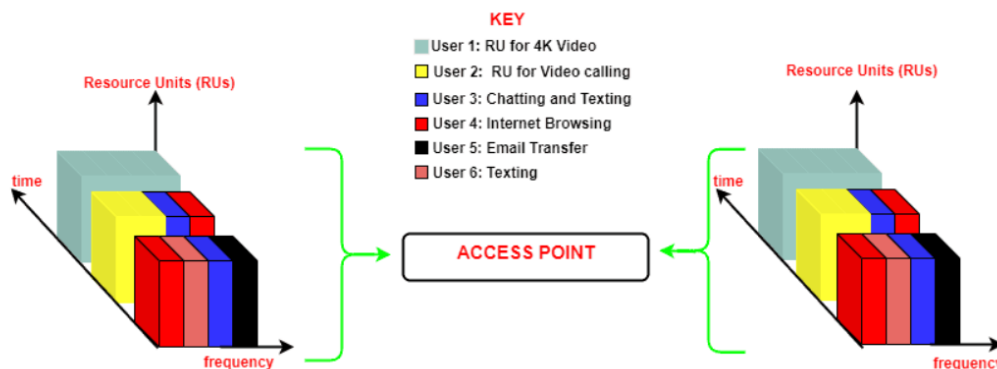
This innovation led to the emergence of the OFDMA protocol. To accomplish this, we utilized the channel resource by combining frequency and time in an orthogonal manner. Now, let's delve into the functionality of the OFDMA protocol:

The OFDMA protocol actively takes a channel and partitions it into smaller frequency units which we know as resource units (RUs). We then allocate these RUs to users according to their respective data consumption demands.

To ensure interference avoidance, we divide the RUs in an orthogonal manner. Subsequently, we employ a single access point (AP) to aggregate all assigned users to their respective RUs, facilitating the delivery of their messages to the intended recipients.

Thanks to the dynamic efficiency we embedded in the design of the OFDMA protocol, we can assign receivers RUs based on the data size they expect to receive.

This is a typical OFDMA protocol layout:



The OFDMA protocol significantly mitigates the unpredictability experienced by users in highly congested areas by enabling more users to connect concurrently to a single access point (AP). In doing so, OFDMA effectively addresses congestion issues.

Furthermore, this protocol exemplifies exceptional efficiency by seamlessly allocating either the entirety of a channel or dividing it to accommodate multiple users, depending on the specific traffic demands.

➤ Design Issues in Network Layer

The Network layer is majorly focused on getting packets from the source to the destination, routing error handling, and congestion control. Before learning about design issues in the network layer, let's learn about its various functions.

- **Addressing:** Maintains the address at the frame header of both source and destination and performs addressing to detect various devices in the network.
- **Packeting:** This is performed by Internet Protocol. The network layer converts the packets from its upper layer.
- **Routing:** It is the most important functionality. The network layer chooses the most relevant and best path for the data transmission from source to destination.

- **Inter-networking:** It works to deliver a logical connection across multiple devices.

Network Layer Design Issues

The network layer comes with some design issues that are described as follows:

1. Store and Forward packet switching

The host sends the packet to the nearest router. This packet is stored there until it has fully arrived once the link is fully processed by verifying the checksum then it is forwarded to the next router till it reaches the destination. This mechanism is called "Store and Forward packet switching."

2. Services provided to the Transport Layer

Through the network/transport layer interface, the network layer transfers its **patterns** services to the transport layer. These services are described below. But before providing these services to the transfer layer, the following goals must be kept in mind:-

- Offering services must not depend on router technology.
- The transport layer needs to be protected from the type, number, and topology of the available router.
- The network addresses for the transport layer should use uniform numbering patterns, also at LAN and WAN connections.

Based on the connections there are 2 types of services provided :

- **Connectionless** - The routing and insertion of packets into the subnet are done individually. No added setup is required.
- **Connection-Oriented** - Subnet must offer reliable service and all the packets must be transmitted over a single route.

3. Implementation of Connectionless Service

Packets are termed as "datagrams" and corresponding subnets as "datagram subnets". When the message size that has to be transmitted is 4 times the size of the packet, then the network layer divides into 4 packets and transmits each packet to the router via. a few protocols. Each data packet has a destination address and is routed independently irrespective of the packets.

4. Implementation of Connection-Oriented service:

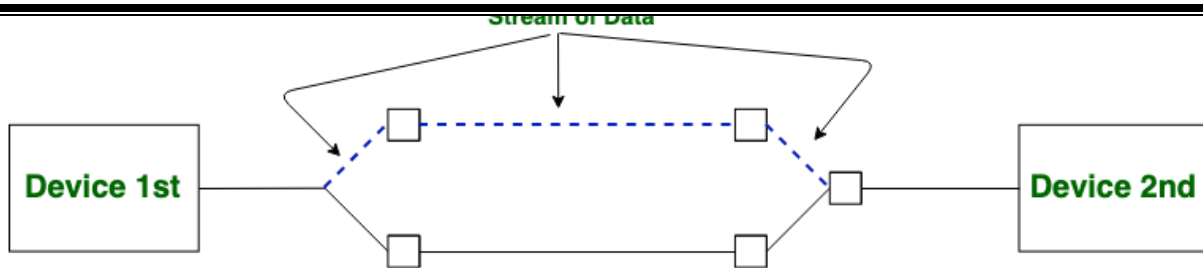
To use a connection-oriented service, first, we establish a connection, use it, and then release it. In connection-oriented services, the data packets are delivered to the receiver in the same order in which they have been sent by the sender. It can be done in either two ways :

- **Circuit Switched Connection** - A dedicated physical path or a circuit is established between the communicating nodes and then the data stream is transferred.
- **Virtual Circuit Switched Connection** - The data stream is transferred over a packet switched network, in such a way that it seems to the user that there is a dedicated path from the sender to the receiver. A virtual path is established here. While, other connections may also be using the same path.

Connection-less vs Connection-Oriented

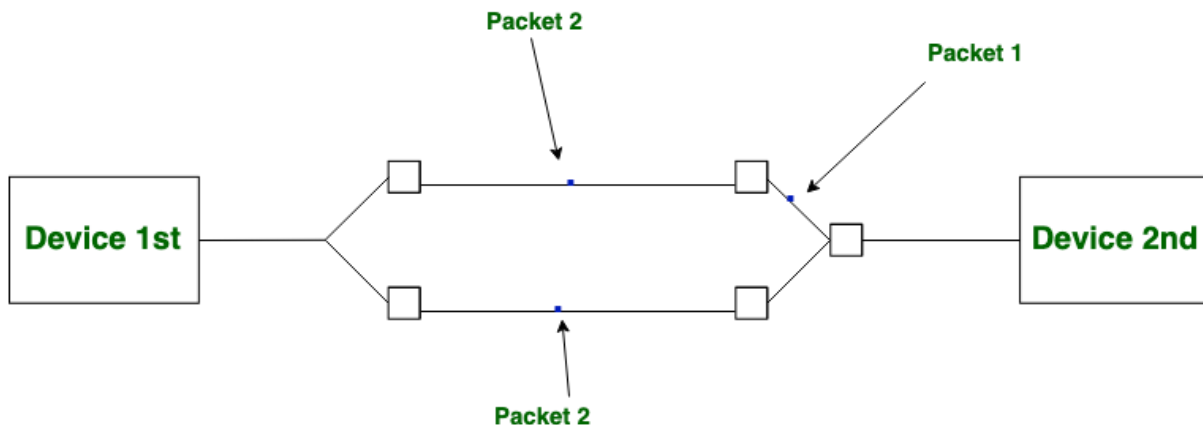
Both Connection-less and Connection-Oriented are used for the connection establishment between two or more devices. These types of services are provided by the [Network Layer](#).

Connection-oriented service: In connection-Oriented service we have to establish a connection between sender and receiver before communication. Handshke method is used to establish a connection between sender and receiver. Connection-Oriented service include both connection establishment as well as connection termination phase. Real life example of this service is telephone service, for conversation we have to first establish a connection.



CONNECTION-ORIENTED SERVICE

Connection-less service: In Connection-Less service no need of connection establishment and connection termination. This Service does not give a guarantee of reliability. In this service, Packets may follow the different path to reach their destination. Real life examples of this service is postal system, Online gaming, real-time video and audio streaming etc.



CONNECTIONLESS SERVICE

➤ Differences Between Virtual Circuits and Datagram Networks

Computer networks that provide connection-oriented services are called Virtual Circuits while those providing connection-less services are called Datagram networks. For prior knowledge, the Internet that we use is based on a Datagram network (connection-less) at the network level as all packets from a source to a destination do not follow the same path.

Virtual Circuits

- It is connection-oriented, meaning that there is a reservation of resources like buffers, [CPU](#), [bandwidth](#), etc. for the time in which the newly set VC is going to be used by a data transfer session.
- The first sent packet reserves resources at each server along the path. Subsequent packets will follow the same path as the first sent packet for the connection time.
- Since all the packets are going to follow the same path, a global header is required. Only the first packet of the connection requires a global header, the remaining packets generally don't require global headers.
- Since all packets follow a specific path, packets are received in order at the destination.
- Virtual Circuit Switching ensures that all packets successfully reach the Destination. No packet will be discarded due to the unavailability of resources.
- From the above points, it can be concluded that [Virtual Circuits](#) are a highly reliable method of data transfer.
- The issue with virtual circuits is that each time a new connection is set up, resources and extra information have to be reserved at every router along the path, which becomes problematic if many clients are trying to reserve a router's resources simultaneously.

- It is used by the [ATM \(Asynchronous Transfer Mode\) Network](#), specifically for Telephone calls.

Types of Virtual Circuit

1. Permanent Virtual Circuits(PVC): The communication management station, which is the telco's central office, manually configures the switches, which offer performance comparable to dedicated lines. The main use for these always-on circuits is high-speed communication. PVCs require telco resources (switches) to be allocated to a single communication circuit whether or not that circuit is in use, making them an expensive solution for wide-area networks (WANs).

2. Switched Virtual circuits (SVCs): As soon as a communication session is established, the switches are set up. SVCs are released at the conclusion of the session and can be used to create new channels of communication. This is the process of normal phone communication. SVCs, which are billed on a per-minute basis, are generally utilised in WANs when backups to dedicated leased lines are required.

Benefits of Virtual Circuit

- The recipient receives the sender's packets in the same order as they were sent.
- A secure network link is called a virtual circuit.
- Overhead is not required for any packet.
- A single global packet overhead is used in a virtual circuit.

Drawbacks of Virtual Circuits

- The cost of implementing a virtual circuit is high.
- It provides only services based on connections.
- In order to transmit, a new link needs to be created permanently.

Datagram Networks

- It is a [connection-less service](#). There is no need for reservation of resources as there is no dedicated path for a connection session.
- All packets are free to use any available path. As a result, intermediate routers calculate routes on the go due to dynamically changing routing tables on routers.
- Since every packet is free to choose any path, all packets must be associated with a header with proper information about the source and the upper layer data.
- The connection-less property makes data packets reach the destination in any order, which means that they can potentially be received out of order at the receiver's end.
- Datagram networks are not as reliable as Virtual Circuits.
- The major drawback of Datagram Packet switching is that a packet can only be forwarded if resources such as the buffer, CPU, and bandwidth are available. Otherwise, the packet will be discarded.
- But it is always easy and cost-efficient to implement datagram networks as there is no extra headache of reserving resources and making a dedicated each time an application has to communicate.
- It is generally used by the IP network, which is used for Data services like the Internet.

Benefits of Datagram Networks

- The flexibility of datagram networks is one of its main benefits.
- They are better at managing network congestion. Datagram networks are able to adjust to variations in network traffic and identify several paths for packets to take in order to reach their intended destination because every packet is handled separately.
- In big and complicated networks in particular, this can lead to decreased latency and increased network performance.

- In addition, datagram networks scale more easily than other kinds of networks. Datagram networks are the ideal option for contemporary communication systems, such as the Internet of Things (IoT) and real-time data streaming applications, due to their scalability.

Drawbacks of Datagram Networks

- The lack of assured delivery in datagram networks is one of their primary disadvantages. There is no assurance that all packets will arrive at their destination or in the right order because they are sent separately.
- Datagram networks also have the drawback of being vulnerable to security breaches. Datagram networks are particularly susceptible to network assaults including spoofing, eavesdropping, and denial of service (DoS) attacks since they don't create a dedicated connection between the sender and the recipient.
- Moreover, datagram networks may not always support guarantees of quality of service (QoS). While certain applications may benefit from QoS capabilities provided by some protocols, like the Real-time Transport Protocol (RTP), datagram networks as a whole do not provide a centralised method for allocating priorities and controlling network traffic.

Difference Between Virtual Circuits and Datagram Networks

Criteria	Virtual Circuit Networks	Datagram Networks
Connection Establishment	Prior to data transmission, a connection is established between sender and receiver.	No connection setup is required.
Routing	Routing decisions are made once during connection setup and remain fixed throughout the duration of the connection.	Routing decisions are made independently for each packet and can vary based on network conditions.
Flow Control	Uses explicit flow control, where the sender adjusts its rate of transmission based on feedback from the receiver.	Uses implicit flow control, where the sender assumes a certain level of available bandwidth and sends packets accordingly.
Congestion Control	Uses end-to-end congestion control, where the sender adjusts its rate of transmission based on feedback from the network.	Uses network-assisted congestion control, where routers monitor network conditions and may drop packets or send congestion signals to the sender.
Error Control	Provides reliable delivery of packets by detecting and retransmitting lost or corrupted packets.	Provides unreliable delivery of packets and does not guarantee delivery or correctness.

Criteria	Virtual Circuit Networks	Datagram Networks
Overhead	Requires less overhead per packet because connection setup and state maintenance are done only once.	Requires more overhead per packet because each packet contains information about its destination address and other routing information.
Example Protocol	ATM, Frame Relay	IP (Internet Protocol)